Single-Variable Calculus

Change, Accumulation, and Approximation

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Single-Variable Calculus: Change, Accumulation, and Approximation.
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Contents

Preface

How to Read This Book

The tag system

Suggested course paths

Part I

The Two Questions of Calculus

Chapter 1

Velocity, Distance, and the First Shape of Calculus

Core

1.1 The dashboard: speedometer and odometer

1.1.1 Distance as accumulated motion

1.1.2 Velocity as changing distance

1.1.3 Units: miles, hours, miles per hour

1.1.4 The first question: if we know distance, can we find velocity?

1.1.5 The second question: if we know velocity, can we find distance?

Core

1.2 Constant velocity: where slope and area first meet

1.2.1 The graph of distance when velocity is constant

1.2.2 Slope of the distance graph

1.2.3 Area under the velocity graph

1.2.4 Why slope and area are inverse clues

Core

1.3 Forward, backward, and signed motion

1.3.1 Positive and negative velocity

1.3.2 Displacement versus distance traveled

1.3.3 Area below the time axis

1.3.4 A first piecewise-defined motion

Bridge

1.4 Calculus without limits: finite differences and finite sums

1.4.1 Difference tables

1.4.2 Accumulated differences telescope

1.4.3 Step velocities and piecewise linear distance

1.4.4 The finite version of the Fundamental Theorem

Core

1.5 Changing velocity and the need for limits

1.5.1 Average velocity over a time interval

1.5.2 Shorter intervals and better estimates

1.5.3 The tangent line problem

1.5.4 The area problem

1.5.5 Why finite arithmetic is no longer enough

Bridge

1.6 A first map of the book

1.6.1 Derivative: from distance to velocity

1.6.2 Integral: from velocity to distance

1.6.3 Approximation: how to work when exact formulas fail

1.6.4 Differential equations: when the law gives the derivative

1.7 Discovery problems and projects

Project

1.7.1 Read a trip from a speed graph

Project

1.7.2 Design two different trips with the same displacement

Bridge

1.7.3 Finite differences of squares and cubes

1.7.4 Hook: a changing slope needs a precise language of nearness

Chapter 2

Numbers, Functions, and Models

Core

2.1 The real line and intervals

2.1.1 Points as numbers

2.1.2 Open, closed, and half-open intervals

2.1.3 Absolute value as distance

2.1.4 Inequalities as a way to control error

Proof / Optional

2.2 Completeness and the idea of no gaps

2.2.1 Rational numbers and the missing square root

2.2.2 Nested intervals

2.2.3 Least upper bound property

2.2.4 Why calculus needs the real numbers

Core

2.3 Functions as assignments

2.3.1 Input, output, domain, and range

2.3.2 Functions from formulas, graphs, tables, and words

2.3.3 Piecewise-defined functions

2.3.4 Difference quotient as a function of two inputs

Core

2.4 Graphs and transformations

2.4.1 Shifts and scalings

2.4.2 Reflections and stretching

2.4.3 Even and odd symmetry

2.4.4 Reading intercepts, sign, and intervals from a graph

Core

2.5 Combining functions

2.5.1 Sums, products, and quotients

2.5.2 Composition

2.5.3 Inverse functions

2.5.4 When an inverse exists

Core

2.6 A catalog of essential functions

2.6.1 Linear functions and units

2.6.2 Powers and polynomials

2.6.3 Rational functions and asymptotic behavior

2.6.4 Algebraic functions

2.6.5 Trigonometric functions in radians

2.6.6 Exponential functions

2.6.7 Logarithms as inverse functions

Core

2.7 Models before calculus

2.7.1 Linear growth

2.7.2 Power-law scaling

2.7.3 Exponential growth and decay

2.7.4 Periodic motion

2.7.5 Choosing a model from units and data

Computing

2.8 Technology and graphs

2.8.1 Windows can lie

2.8.2 Sampling can miss behavior

2.8.3 Numerical evidence versus proof

2.8.4 A first warning about roundoff

2.9 Chapter review and discovery problems

2.9.1 Concept check: function versus formula

Project

2.9.2 Skills: graph transformations and domains

2.9.3 Project: fit a simple model to data

2.9.4 Hook: limits make “near” into mathematics

Part II

Limits, Continuity, and the Derivative

Chapter 3

Limits and Continuity

Core

3.1 Limits from motion and approximation

3.1.1 Average velocity as the interval shrinks

3.1.2 Decimal approximations to a number

3.1.3 The informal meaning of a limit

3.1.4 Limits from graphs and tables

Core

3.2 Calculating limits

3.2.1 Limit laws

3.2.2 Polynomials and rational functions

3.2.3 Removing a removable discontinuity

3.2.4 The squeeze idea

Core

3.3 One-sided limits and infinite behavior

3.3.1 Left-hand and right-hand limits

3.3.2 Infinite limits and vertical asymptotes

3.3.3 Limits at infinity

3.3.4 Horizontal and slant asymptotes

Proof

3.4 The precise definition of a limit

3.4.1 The epsilon challenge

3.4.2 Delta as a response

3.4.3 Proving simple limits

3.4.4 Why the definition matches the picture

Core

3.5 Continuity

3.5.1 Continuity at a point

3.5.2 Continuity on an interval

3.5.3 Algebra of continuous functions

3.5.4 Continuity of compositions

3.5.5 Removable, jump, and infinite discontinuities

Core / Proof

3.6 Two theorems about continuous functions

3.6.1 Intermediate Value Theorem

3.6.2 Bisection as a constructive proof idea

3.6.3 Extreme Value Theorem

3.6.4 Why closed and bounded intervals matter

Counterexample

3.7 Warning examples

3.7.1 A function with no limit at a jump

3.7.2 A limit that is infinite, not finite

3.7.3 IVT fails without continuity

3.7.4 EVT fails on an open interval

3.7.5 EVT fails on an unbounded interval

Bridge

3.8 Sequences as limits indexed by whole numbers

3.8.1 Sequence notation

3.8.2 Convergence and divergence

3.8.3 Monotone bounded sequences

3.8.4 Sequences as preparation for infinite series

3.9 Chapter review and discovery problems

Proof

3.9.1 Skills: compute and interpret limits

Computing / Project

3.9.2 Proof practice with epsilon and delta

3.9.3 Project: bisection method for a root

3.9.4 Hook: a tangent line is a limit of secant lines

Chapter 4

The Derivative

4.1 Average rate of change

Core

4.1.1 Secant slope

4.1.2 Average velocity

4.1.3 Average rate from data

4.1.4 Units of a rate

4.2 Instantaneous rate of change

Core

4.2.1 Shrinking the time interval

4.2.2 The derivative at a point

4.2.3 Tangent slope

4.2.4 The derivative as a limit

4.3 The derivative as a function

Core

4.3.1 From one tangent slope to many

4.3.2 Notations: $f'(x)$, dy/dx, $D_x f$

4.3.3 Reading f' from the graph of f

4.3.4 Reading f from the graph of f'

4.4 Local linear approximation

Core / Bridge

4.4.1 The tangent line as a local model

4.4.2 $f(a+h) \approx f(a) + f'(a)h$

4.4.3 Differentials $dy = f'(x)\,dx$

4.4.4 Sensitivity and propagated error

4.5 Differentiability and continuity

Core / Proof

4.5.1 Differentiability implies continuity

4.5.2 Why the converse is false

4.5.3 Corners, cusps, vertical tangents, and jumps

4.5.4 Absolute value as the first warning example

Core

4.6 First derivative formulas

4.6.1 Constant and identity functions

4.6.2 Power functions with positive integer powers

4.6.3 Polynomials

4.6.4 Constant multiples and sums, introduced by examples

Core

4.7 Motion revisited

4.7.1 Position, velocity, and acceleration

4.7.2 Speed versus velocity

4.7.3 Falling under constant acceleration

4.7.4 Units and signs in motion problems

Counterexample

4.8 Warning examples

Optional

4.8.1 A continuous function with no derivative at one point

Computing

4.8.2 A derivative that exists but is not continuous

4.8.3 A graphing calculator misses a sharp corner

4.9 Chapter review and discovery problems

4.9.1 Skills: derivative from definition

4.9.2 Concept check: tangent line versus secant line

Computing / Project

4.9.3 Project: estimate velocity from noisy position data

4.9.4 Hook: computing every derivative from the definition is too slow

Chapter 5

Differentiation Rules and Elementary Functions

Core

5.1 The algebra of derivatives

5.1.1 Constant multiple rule

5.1.2 Sum and difference rules

5.1.3 Why derivative rules are linear

5.1.4 Derivatives of polynomials

Core

5.2 Product and quotient rules

5.2.1 Why the product rule has two terms

5.2.2 Product rule from changing area

5.2.3 Reciprocal rule

5.2.4 Quotient rule

Core

5.3 The chain rule

5.3.1 A temperature that changes because position changes

5.3.2 Composition as change inside change

5.3.3 The chain rule formula

5.3.4 Tree diagrams for nested functions

5.3.5 Chain rule in differential notation

Core

5.4 Implicit differentiation

5.4.1 Curves not solved for y

5.4.2 Differentiating both sides

5.4.3 Tangent lines to implicit curves

5.4.4 Higher derivatives implicitly

Core

5.5 Derivatives of trigonometric functions

5.5.1 Radians and the limit $\sin h/h \to 1$

5.5.2 Derivative of sine

5.5.3 Derivative of cosine

5.5.4 Tangent, secant, cosecant, and cotangent

5.5.5 Simple harmonic motion preview

Core

5.6 Derivatives of exponential and logarithmic functions

5.6.1 Exponential growth as a rate law

5.6.2 The special base e

5.6.3 Derivative of e^x

5.6.4 Derivative of a^x

5.6.5 Derivative of $\ln x$

5.6.6 Logarithmic differentiation

Core

5.7 Inverse functions and inverse trigonometric functions

5.7.1 Derivative of an inverse function

5.7.2 Why nonzero derivative matters

5.7.3 $\arcsin x$, $\arctan x$, and friends

5.7.4 Inverse trigonometric derivatives in integration preview

Core / Optional

5.8 General power rule and hyperbolic functions

5.8.1 Derivative of x^r

5.8.2 Hyperbolic sine and cosine

5.8.3 Hyperbolic identities

Optional

5.8.4 Why these functions appear in hanging cables

Counterexample

5.9 Warning examples

5.9.1 The product rule is not $(fg)' = f'g'$

5.9.2 The chain rule needs composition order

5.9.3 An inverse can fail to be differentiable when the tangent is horizontal

5.10 Chapter review and discovery problems

5.10.1 Skills: mixed derivative calculations

Proof

5.10.2 Proof practice: product and chain rules

Computing / Project

5.10.3 Project: derivative formulas from numerical experiments

5.10.4 Hook: derivatives compute slopes, but slopes tell the shape of a graph

Part III

What Derivatives Reveal

Chapter 6

Shape, Extremes, and the Mean Value Theorem

Core

6.1 Increasing and decreasing functions

6.1.1 Reading motion from the sign of velocity

6.1.2 First derivative sign charts

6.1.3 Intervals of increase and decrease

6.1.4 Critical numbers

Core

6.2 Local and absolute extrema

6.2.1 Local maximum and local minimum

6.2.2 Fermat's theorem

6.2.3 Closed interval method

6.2.4 Endpoints matter

Core / Proof

6.3 Rolle's Theorem and the Mean Value Theorem

6.3.1 A road trip with average speed

6.3.2 Rolle's Theorem

6.3.3 Mean Value Theorem

6.3.4 Consequences: constant derivative and equal derivatives

Core

6.4 Concavity and second derivatives

6.4.1 Velocity changing: acceleration

6.4.2 Concave up and concave down

6.4.3 Inflection points

6.4.4 Second derivative test

Core

6.5 Graph sketching with calculus

6.5.1 Sign charts for f' and f''

6.5.2 Asymptotes and end behavior

Computing

6.5.3 A complete curve-sketching workflow

6.5.4 When technology helps and when it misleads

Core

6.6 Indeterminate forms and l'Hospital's Rule

6.6.1 The forms $0/0$ and ∞/∞

6.6.2 Why derivatives can reveal a limiting ratio

6.6.3 Other indeterminate forms

6.6.4 Comparing growth rates

Counterexample

6.7 Warning examples

6.7.1 Fermat's theorem fails at an endpoint

6.7.2 MVT fails without continuity

6.7.3 MVT fails without differentiability

6.7.4 Second derivative test can be inconclusive

6.7.5 l'Hospital's Rule cannot be used on every quotient

6.8 Chapter review and discovery problems

Proof

6.8.1 Skills: graph from derivatives

Project

6.8.2 Proof practice with MVT

6.8.3 Project: reconstruct a trip from acceleration data

6.8.4 Hook: knowing shape is useful, but many problems ask for the best possible choice

Chapter 7

Optimization, Related Rates, and Models

7.1 Related rates

Core

7.1.1 Quantities changing together

7.1.2 Differentiating a relation with respect to time

7.1.3 Geometry problems: ladders, shadows, cones

7.1.4 Units as a guardrail

7.2 Optimization problems

Core

7.2.1 Translating a word problem into a function

7.2.2 Constraints in one variable

7.2.3 Geometry: boxes, fences, cans, and light paths

7.2.4 Endpoint checks and physical feasibility

7.3 Linear approximation in applications

Core

7.3.1 Estimating hard numbers quickly

7.3.2 Error propagation

7.3.3 Relative and percentage error

7.3.4 Sensitivity in measurement

7.4 Newton's method

Core / Computing

7.4.1 Tangent lines as root finders

7.4.2 The iteration formula

7.4.3 Convergence when the starting point is good

7.4.4 Failure modes: cycles and bad tangents

7.5 Exponential growth and decay

Core

7.5.1 The differential equation $y' = ky$

7.5.2 Half-life and doubling time

7.5.3 Continuously compounded interest

7.5.4 Newton's law of cooling

Core / Bridge

7.6 Logistic growth and limited resources

7.6.1 Why pure exponential growth cannot last

7.6.2 The logistic differential equation

7.6.3 Equilibria and stability from a graph

7.6.4 Population and spread models

Optional / Project

7.7 Light, time, and extremal principles

7.7.1 Reflection from shortest path

7.7.2 Refraction and Snell's law

7.7.3 The seed of the calculus of variations

Counterexample

7.8 Warning examples

7.8.1 A critical point outside the physical domain

7.8.2 A local maximum that is not the absolute maximum

7.8.3 Newton's method converges to the wrong root or does not converge

7.9 Chapter review and applications

Project

7.9.1 Skills: related rates and optimization

Computing / Project

7.9.2 Project: design a least-material container

7.9.3 Project: model cooling data

7.9.4 Hook: derivatives measure change; now we need to add change back up

Part IV

Integrals and the Fundamental Theorem

Chapter 8

The Integral as Accumulation

Core

8.1 Distance from velocity

8.1.1 Constant velocity revisited

8.1.2 Piecewise constant velocity

8.1.3 Estimating distance from sampled speeds

8.1.4 Signed displacement versus total distance

Core

8.2 Area under a graph

8.2.1 Rectangular estimates

8.2.2 Left, right, and midpoint sums

8.2.3 Upper and lower sums

8.2.4 Area as a limiting process

Core

8.3 Sigma notation and finite sums

8.3.1 Summation notation

8.3.2 Arithmetic and geometric sums

8.3.3 Sums of powers

8.3.4 From finite sums to integrals

Core

8.4 The definite integral

8.4.1 Partitions and sample points

8.4.2 Riemann sums

8.4.3 Definition of the definite integral

8.4.4 Integrability, informally first

Core

8.5 Signed area and net change

8.5.1 Area above the axis

8.5.2 Area below the axis

8.5.3 Cancellation and meaning

8.5.4 Total accumulation from absolute value

Core

8.6 Properties of the integral

8.6.1 Linearity

8.6.2 Additivity over intervals

8.6.3 Order properties

8.6.4 Average value of a function

8.6.5 Symmetry

Proof / Optional

8.7 Existence of integrals

8.7.1 Continuous functions are integrable

8.7.2 Monotone functions are integrable

Counterexample

8.7.3 Functions with finitely many jump discontinuities

8.7.4 A bounded function that is not Riemann integrable

8.8 Chapter review and discovery problems

Project

8.8.1 Skills: set up Riemann sums

Computing

8.8.2 Project: estimate energy use from power data

8.8.3 Computing: compare left, right, midpoint, and trapezoid estimates

8.8.4 Hook: an accumulation function has a slope, and that slope is surprisingly simple

Chapter 9

The Fundamental Theorem of Calculus

Core

9.1 Accumulation functions

9.1.1 Area with a moving endpoint

9.1.2 $A(x) = \int_a^x f(t)\,dt$

9.1.3 Estimating the change in an accumulation function

9.1.4 The slope of accumulated area

Core / Proof

9.2 Fundamental Theorem, Part I

9.2.1 The derivative of an accumulation function

9.2.2 Continuity as the needed hypothesis

9.2.3 Geometric proof idea

9.2.4 Examples with variable upper limits

Core

9.3 Fundamental Theorem, Part II

9.3.1 Antiderivatives evaluate definite integrals

9.3.2 Net change theorem

9.3.3 Position from velocity

9.3.4 Accumulated rate in science and economics

Core

9.4 Indefinite integrals

9.4.1 Antiderivative families

9.4.2 The constant of integration

9.4.3 Initial value problems

9.4.4 Basic antiderivative formulas

Core

9.5 Substitution

9.5.1 Reversing the chain rule

9.5.2 Changing variables in a definite integral

9.5.3 Orientation and reversed limits

9.5.4 Substitution in motion and geometry

Bridge

9.6 Differential notation and oriented length

9.6.1 dx as an oriented step

9.6.2 $f(x)\,dx$ as the thing being accumulated

9.6.3 Pulling an integrand through $x = g(u)$, without abstraction

9.6.4 The one-dimensional seed of differential forms

Counterexample

9.7 Warning examples

9.7.1 A discontinuity and the derivative of an accumulation function

9.7.2 Forgetting the new limits in substitution

9.7.3 Treating $\int f(x)\,dx$ like multiplication

9.7.4 Confusing area with net signed area

9.8 Chapter review and discovery problems

9.8.1 Skills: FTC and substitution

Proof

9.8.2 Proof practice: FTC Part I in a simple case

Project

9.8.3 Project: reconstruct a function from its rate

9.8.4 Hook: once integration is computable, it can measure more than area

Part V

Applications and Techniques of Integration

Chapter 10

What Integrals Measure

Core

10.1 Area between curves

10.1.1 Top minus bottom

10.1.2 Right minus left

10.1.3 Splitting at intersections

10.1.4 Area from data and graphs

Core

10.2 Volumes by slicing

10.2.1 Cross-sectional area as a function

10.2.2 Solids of revolution: disks and washers

10.2.3 Non-circular cross sections

10.2.4 Choosing the slicing direction

Core

10.3 Cylindrical shells

10.3.1 A thin shell unwrapped

10.3.2 Shell method formula

10.3.3 When shells beat washers

10.3.4 Comparing two methods on the same solid

Core

10.4 Mass, density, and center of mass

10.4.1 Linear density and mass of a rod

10.4.2 Moments

10.4.3 Center of mass on a line

Bridge

10.4.4 Lamina preview by slices

Core

10.5 Work and energy

10.5.1 Constant force and variable force

10.5.2 Springs and Hooke's law

10.5.3 Lifting a chain

10.5.4 Pumping fluid from a tank

Core / Optional

10.6 Hydrostatic force

10.6.1 Pressure depends on depth

10.6.2 Force on a vertical plate

10.6.3 Choosing strips

10.6.4 Dams and windows

Core

10.7 Average value and probability

10.7.1 Average value of a continuous function

10.7.2 Probability density

10.7.3 Expected value

10.7.4 Normal distribution as a preview of non-elementary integrals

Core

10.8 Arc length and surface area of revolution

10.8.1 Arc length from small straight pieces

10.8.2 Arc length of a graph

10.8.3 Surface area by rotating a curve

10.8.4 Why some natural lengths have no elementary antiderivative

10.9 Chapter review and applications

Project

10.9.1 Skills: choose the small piece

Project

10.9.2 Project: compare volume methods

10.9.3 Project: work needed to pump out a real tank shape

10.9.4 Hook: many integrals cannot be found by substitution alone

Chapter 11

Techniques of Integration

11.1 Integration by parts

Core

11.1.1 Reversing the product rule

11.1.2 Choosing u and dv

11.1.3 Repeated integration by parts

11.1.4 Definite integrals and boundary terms

11.2 Trigonometric integrals

Core

11.2.1 Powers of sine and cosine

11.2.2 Powers of tangent and secant

11.2.3 Product-to-sum identities

11.2.4 Strategy over memorization

11.3 Trigonometric substitution

Core

11.3.1 Square roots from circles and triangles

11.3.2 $a^2 - x^2$, $a^2 + x^2$, and $x^2 - a^2$

11.3.3 Returning to the original variable

11.3.4 Hyperbolic substitutions

Optional

11.4 Rational functions and partial fractions

Core

11.4.1 Polynomial division

11.4.2 Distinct linear factors

11.4.3 Repeated linear factors

11.4.4 Irreducible quadratic factors

11.4.5 Completing the square

11.5 Strategy for integration

Core

11.5.1 Recognizing the structure of an integrand

11.5.2 Substitution first or parts first?

11.5.3 Algebraic simplification

11.5.4 When no elementary antiderivative exists

Computing

11.6 Computer algebra systems and integral tables

11.6.1 Checking without surrendering thought

11.6.2 Equivalent antiderivatives can look different

11.6.3 Branch and domain issues

11.6.4 Special functions born from integrals

Core

11.7 Improper integrals

11.7.1 Infinite intervals

11.7.2 Infinite discontinuities

11.7.3 Convergence and divergence

11.7.4 Comparison tests for improper integrals

11.7.5 The p-integrals

Counterexample

11.8 Warning examples

11.8.1 An antiderivative exists formally but the improper integral diverges

11.8.2 Cancellation hides divergence

11.8.3 A CAS gives an antiderivative with the wrong domain

11.8.4 A trigonometric substitution triangle used outside its range

11.9 Chapter review and discovery problems

Project

11.9.1 Skills: choose and apply integration techniques

Computing / Project

11.9.2 Project: build an integration decision tree

11.9.3 Project: compare human and CAS antiderivatives

11.9.4 Hook: exact integration is only one way to compute an accumulated amount

Chapter 12

Numerical Integration, Error, and Computation

12.1 Why approximate integrals?

Core / Computing

12.1.1 Data instead of formulas

12.1.2 Non-elementary antiderivatives

12.1.3 Speed versus accuracy

12.1.4 The role of error estimates

12.2 Midpoint and trapezoidal rules

Core

12.2.1 Geometry of each rule

12.2.2 Composite rules

12.2.3 Error patterns from concavity

12.2.4 Error bounds

12.3 Simpson's Rule

Core

12.3.1 Quadratic approximation on pairs of intervals

12.3.2 Composite Simpson's Rule

12.3.3 Why Simpson's Rule is often much better

12.3.4 Simpson error bound

12.4 Numerical differentiation

Computing / Optional

12.4.1 Forward, backward, and centered differences

12.4.2 Truncation error

12.4.3 Roundoff error

12.4.4 Why smaller h is not always better

12.5 Newton's method revisited

Computing

12.5.1 Stopping criteria

12.5.2 Error estimates

Optional

12.5.3 Multiple roots

Project

12.5.4 A first glimpse of chaotic iteration

12.6 Computing experiments

12.6.1 Approximate π by integration

12.6.2 Estimate a probability from a density

12.6.3 Compare rules on smooth and rough functions

12.6.4 Build a small numerical-integration program

12.7 Chapter review and discovery problems

12.7.1 Skills: numerical rules and error bounds

12.7.2 Concept check: exact value versus approximation

12.7.3 Hook: a differential equation often cannot be solved exactly either

Part VI

Differential Equations and Curves

Chapter 13

Differential Equations: Laws Written as Derivatives

13.1 Modeling with differential equations

Core

13.1.1 A law that gives a rate

13.1.2 Solution as an unknown function

13.1.3 Initial conditions

13.1.4 Checking a proposed solution

13.2 Direction fields

Core

13.2.1 Slope field as a picture of a differential equation

13.2.2 Equilibrium solutions

13.2.3 Qualitative behavior

13.2.4 Reading long-term behavior from the field

13.3 Euler's method

Core / Computing

13.3.1 Following the slope field step by step

13.3.2 Step size and accumulated error

13.3.3 Improved Euler method

Optional

13.3.4 Numerical solutions as approximate functions

13.4 Separable equations

Core

13.4.1 Separating variables

13.4.2 Growth and decay revisited

13.4.3 Cooling and mixing

13.4.4 Implicit solutions

13.5 Logistic and threshold models

Core

13.5.1 Carrying capacity

13.5.2 Phase line

Optional

13.5.3 Stable and unstable equilibria

13.5.4 Harvesting and tipping points

Core

13.6 First-order linear equations

13.6.1 Standard form

13.6.2 Integrating factors

13.6.3 Mixing and finance examples

13.6.4 Why the integrating factor works

Optional / Bridge

13.7 Second-order equations and vibration

13.7.1 Simple harmonic motion

13.7.2 The equation $y'' + \omega^2 y = 0$

13.7.3 Damping

13.7.4 Driven oscillation and resonance

Optional

13.8 Systems and interaction models

13.8.1 Two populations

13.8.2 Predator-prey equations as a picture

13.8.3 Why systems lead naturally to multivariable calculus

Counterexample

13.9 Warning examples

13.9.1 A solution lost by dividing by zero

13.9.2 Euler's method follows the wrong behavior with a large step

13.9.3 Same differential equation, different initial conditions

13.10 Chapter review and applications

13.10.1 Skills: solve and interpret first-order equations

Project

13.10.2 Project: model a cooling cup of coffee

Computing / Project

13.10.3 Project: compare Euler approximations with exact solutions

13.10.4 Hook: some curves are easier to describe by how they are traced than by $y = f(x)$

Chapter 14

Parametric Curves, Polar Coordinates, and Conics

14.1 Parametric curves

Core

14.1.1 A curve as a moving point

14.1.2 Eliminating the parameter when possible

14.1.3 Different parametrizations of the same curve

14.1.4 Orientation and speed along a curve

14.2 Calculus with parametric curves

Core

14.2.1 Slope dy/dx from dx/dt and dy/dt

14.2.2 Horizontal and vertical tangents

14.2.3 Concavity

14.2.4 Motion in the plane as a bridge to vectors

14.3 Length and area for parametric curves

Core

14.3.1 Arc length from speed

14.3.2 Surface area from a parametrized curve of revolution

14.3.3 Area enclosed by a parametric curve

14.3.4 Cycloids and rolling motion

14.4 Polar coordinates

Core

14.4.1 Distance and angle as coordinates

14.4.2 Converting between polar and Cartesian coordinates

14.4.3 Polar graphs

14.4.4 Circles, spirals, roses, and limacons

14.5 Calculus in polar coordinates

Core

14.5.1 Slope of a polar curve

14.5.2 Area in polar coordinates

14.5.3 Arc length in polar coordinates

14.5.4 Intersections and repeated tracing

Core

14.6 Conic sections

14.6.1 Parabolas, ellipses, and hyperbolas from focus and directrix

14.6.2 Standard equations

14.6.3 Polar equations of conics

14.6.4 Eccentricity

Optional / Project

14.7 Planetary motion as a capstone

14.7.1 Ellipses in polar form

14.7.2 Kepler's second law as swept area

14.7.3 Central force preview

14.7.4 Why this belongs to multivariable calculus next

Bridge / Optional

14.8 Complex numbers and polar form

14.8.1 Complex numbers as points in the plane

14.8.2 Multiplication as scaling and rotation

14.8.3 Euler's formula preview

14.8.4 Oscillation written exponentially

Counterexample

14.9 Warning examples

14.9.1 A parametrization stops even when the curve continues

14.9.2 A polar curve traced twice

14.9.3 A tangent formula fails when both derivatives vanish

14.10 Chapter review and projects

14.10.1 Skills: parametric and polar calculus

Computing / Project

14.10.2 Project: design a cycloid animation

Project

14.10.3 Project: compare polar and Cartesian descriptions of conics

14.10.4 Hook: infinite processes return, now as sums rather than areas

Part VII

Infinite Processes and Approximation by Polynomials

Chapter 15

Sequences and Infinite Series

15.1 Sequences

Core

15.1.1 Lists with a rule

15.1.2 Convergence and divergence

15.1.3 Monotone bounded sequences

15.1.4 Recursive sequences and fixed points

15.2 Infinite series

Core

15.2.1 Partial sums

15.2.2 Series convergence as a sequence problem

15.2.3 Geometric series

15.2.4 Telescoping series

15.3 The harmonic series and first warnings

Core

15.3.1 Terms go to zero but the series diverges

15.3.2 Necessary condition for convergence

15.3.3 The harmonic series

15.3.4 Grouping terms to see divergence

15.4 Positive series tests

Core

15.4.1 Integral test

15.4.2 Remainder estimates from the integral test

15.4.3 Direct comparison

15.4.4 Limit comparison

15.4.5 p-series

15.5 Alternating series

Core

15.5.1 Alternating Series Test

15.5.2 Alternating Series Estimation Theorem

15.5.3 Conditional convergence

15.5.4 Error as the first omitted term

Core

15.6 Absolute convergence and stronger tests

15.6.1 Absolute versus conditional convergence

15.6.2 Ratio Test

15.6.3 Root Test

15.6.4 Strategy for testing series

Optional / Proof

15.7 Rearrangement and the danger of infinity

15.7.1 Finite sums can be rearranged freely

15.7.2 Infinite conditionally convergent sums cannot

15.7.3 Riemann rearrangement idea

15.7.4 Why absolute convergence is safer

Counterexample

15.8 Warning examples

15.8.1 Terms tend to zero but the series diverges

15.8.2 Ratio Test gives no information

15.8.3 Integral Test used on a nonpositive function

15.8.4 Alternating Series Test used without decreasing terms

15.9 Chapter review and discovery problems

Project

15.9.1 Skills: choose series tests

Computing

15.9.2 Project: estimate sums with rigorous error bounds

15.9.3 Computing: numerical partial sums can mislead

15.9.4 Hook: some infinite sums are functions, not just numbers

Chapter 16

Power Series, Taylor Polynomials, and Taylor Series

Core

16.1 Power series

16.1.1 A polynomial with infinitely many terms

16.1.2 Center of a power series

16.1.3 Radius and interval of convergence

16.1.4 Endpoint testing

Core

16.2 Working with power series

16.2.1 Algebra of power series

16.2.2 Differentiating power series

16.2.3 Integrating power series

16.2.4 Why power series behave better than arbitrary series

Core

16.3 Representing functions by power series

16.3.1 Geometric series as the seed

16.3.2 Series for $1/(1-x)$

16.3.3 Series for $\ln(1+x)$

16.3.4 Series for $\arctan x$

16.3.5 Estimating π from a series

Core

16.4 Taylor polynomials

16.4.1 Matching value and derivatives

16.4.2 Linear, quadratic, and cubic approximations

16.4.3 Taylor polynomial centered at a

16.4.4 Maclaurin polynomials

Core / Proof

16.5 Taylor's theorem and remainders

16.5.1 The error term matters

Optional

16.5.2 Lagrange form of the remainder

16.5.3 Integral form of the remainder

16.5.4 Alternating-series remainders as a special case

Core

16.6 Taylor series for elementary functions

16.6.1 e^x

16.6.2 $\sin x$ and $\cos x$

16.6.3 $\ln(1+x)$

16.6.4 $\arctan x$

16.6.5 Binomial series

Core

16.7 Applications of Taylor series

16.7.1 Estimating function values

16.7.2 Evaluating limits

16.7.3 Approximating integrals

16.7.4 Solving differential equations by series

16.7.5 Error bounds in applications

Optional / Bridge

16.8 Uniform convergence preview

16.8.1 Pointwise convergence of functions

16.8.2 Uniform convergence as one error bound for all points

16.8.3 Interchanging limits and integrals

16.8.4 Why power series are trustworthy inside their radius

Counterexample

16.9 Warning examples

Optional

16.9.1 Taylor series converges but not to the function

16.9.2 Endpoint behavior differs from interior behavior

Optional

16.9.3 Termwise operations fail for arbitrary function series

16.9.4 A good local approximation becomes bad far away

16.10 Chapter review and projects

16.10.1 Skills: power series and Taylor series

16.10.2 Project: build a sine calculator with error control

Computing / Project

16.10.3 Project: compare Taylor approximations visually

Computing / Project

16.10.4 Hook: the series for sine and cosine hide a rotating vector

Part VIII

Capstones and Bridges

Chapter 17

Complex Numbers, Fourier Ideas, and the Road Ahead

17.1 Complex numbers and Euler's formula

Bridge / Optional

17.1.1 Complex plane

17.1.2 Polar form

17.1.3 Multiplication as rotation and scaling

17.1.4 Euler's formula from Taylor series

17.1.5 Why e^{it} is the natural language of oscillation

17.2 Vibrations and second-order equations

Optional

17.2.1 The spring equation

17.2.2 Sine and cosine as natural motions

17.2.3 Energy conservation

17.2.4 Damping and forcing preview

17.3 Fourier series preview

Optional

17.3.1 Building a periodic function from waves

17.3.2 Orthogonality of sine and cosine, geometrically

17.3.3 Square waves and Gibbs phenomenon

17.3.4 Why Fourier series belong after Taylor series

17.4 The one-dimensional Stokes theorem

Bridge

17.4.1 Boundary of an interval

17.4.2 $\int_a^b F'(x)\,dx = F(b) - F(a)$

17.4.3 The FTC as a boundary theorem

17.4.4 Preview of line integrals and differential forms

Optional / Bridge

17.5 Conservation laws in one dimension

17.5.1 Density and flux on a line

17.5.2 Accumulation inside an interval

17.5.3 What crosses the boundary changes what is inside

17.5.4 Preview of divergence in multivariable calculus

Optional / Project

17.6 Least action and the shape of a path

17.6.1 From optimization of numbers to optimization of functions

17.6.2 The brachistochrone story

17.6.3 A gentle first variation calculation

17.6.4 Why this is beyond ordinary single-variable calculus

17.7 Final projects

Project

17.7.1 Model, solve, and test a cooling law

Project

17.7.2 Approximate a difficult integral three ways

Computing / Project

17.7.3 Build a Taylor-series calculator

Project

17.7.4 Analyze a parametric curve from motion data

Bridge / Project

17.7.5 Explain the FTC as the first boundary theorem

Appendix A

Algebra and Inequalities Review

A.1 Factoring and completing the square

A.2 Rational expressions

A.3 Exponents and radicals

A.4 Inequalities and absolute values

A.5 Sigma notation practice

Appendix B

Coordinate Geometry

B.1 Lines

B.2 Circles

B.3 Parabolas, ellipses, and hyperbolas

B.4 Distance and midpoint formulas

B.5 Scaling and shifting graphs

Appendix C

Trigonometry in Radians

C.1 Unit circle definitions

C.2 Basic identities

C.3 Addition formulas

C.4 Inverse trigonometric functions

C.5 Trigonometric equations

Appendix D

Proofs and Completeness

D.1 Least upper bound property

D.2 Monotone convergence theorem

D.3 Bolzano's theorem and bisection

D.4 Extreme Value Theorem proof outline

D.5 Uniform continuity on closed intervals

Appendix E

Technology Notes

E.1 Graphing windows

E.2 Numerical precision and roundoff

E.3 Calculator and CAS syntax

E.4 Simple Python experiments for calculus

E.5 Plotting sequences, sums, and Taylor polynomials

Appendix F

Tables

F.1 Derivative formulas

F.2 Integral formulas

F.3 Common Taylor series

F.4 Series tests

F.5 Numerical integration rules